



DJS Skylark Project

Website Content Package for Portfolio Implementation

SAE Aero Design East
2022

Fort Worth, Texas

Global Top 3 Overall



Competition team with aircraft and awards at SAE Aero Design East 2022.

This PDF is intended as a source document for building the DJS Skylark section of Arya Raut's portfolio website. It combines concise website-ready copy, verified competition achievements, personal contribution bullets, and image placement guidance.

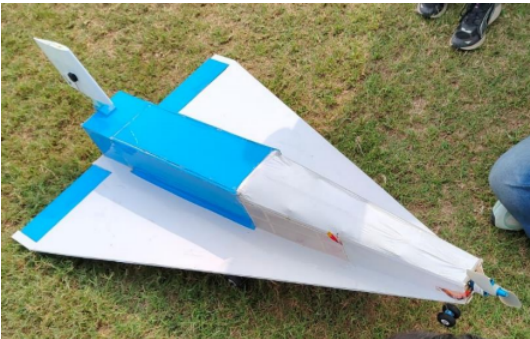
Recommended website positioning: A high-impact aerospace engineering project showing leadership, structural design ownership, Python-based optimization, hands-on aircraft assembly, and international competition performance.

1. Project Overview

DJS Skylark is the official Aero-Design team of DJSCE. The team researches, designs, analyzes, fabricates, and competes with radio-controlled aircraft on an international platform against universities from around the world.

In 2022, DJS Skylark participated in SAE Aero Design East in Fort Worth, Texas, competing in both Regular Class and Micro Class with 42 teams from around the globe.

Portfolio theme: aircraft design, structural engineering, stability analysis, optimization, team leadership, and competition execution.



Prototype aircraft on the competition field.

Quick Facts for Website

Competition	SAE Aero Design East 2022
Location	Fort Worth, Texas
Dates	May 20-22, 2022
Team	DJS Skylark, DJSCE
Classes	Regular Class and Micro Class
Global field	42 teams from around the world
Arya's role	Structure & Stability team lead; aircraft structural design, stability, assembly, repair, troubleshooting, and Python optimization support

2. Competition Achievements - 2022

Use these as achievement cards, counters, or award badges on the website.

Category	Micro Class	Regular Class
Design Report	1st Worldwide	3rd Worldwide
Mission Performance	3rd Worldwide	3rd Worldwide
Technical Presentation	5th Worldwide	4th Worldwide
Overall	3rd Worldwide	3rd Worldwide



Award plaques displayed over the Indian flag, including Micro Class Design Report 1st Place and overall/mission performance awards.

Website-ready achievement summary: DJS Skylark placed 3rd Overall Worldwide in both Regular Class and Micro Class at SAE Aero Design East 2022. The team also earned 1st Worldwide in Micro Class Design Report, 3rd Worldwide in Mission Performance for both classes, and top-five presentation placements.

3. Arya Raut's Contributions

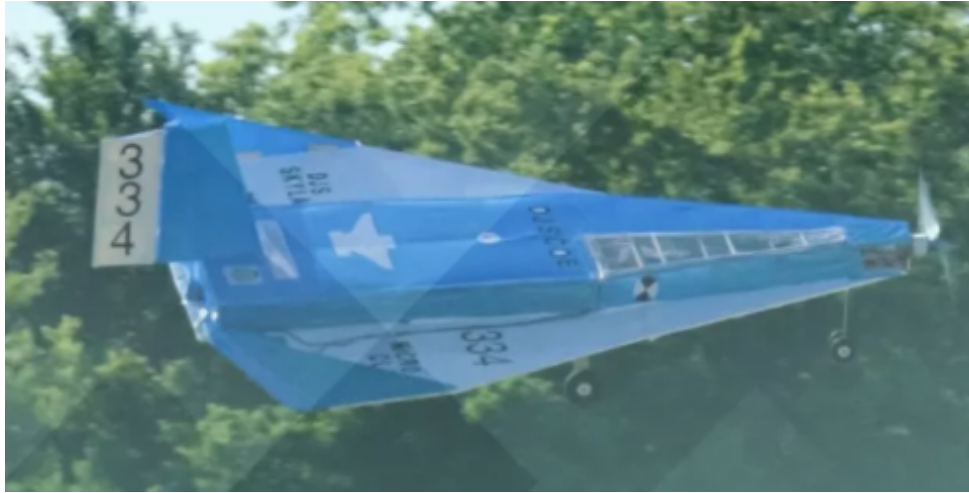
Use this section to connect the team achievement to Arya's personal engineering contribution.

Leadership	Led the Structure & Stability team and mentored six members through design reviews, aircraft build planning, assembly support, and performance troubleshooting.
Structural Design	Drove aircraft structural design and improvement initiatives focused on airframe stability, reliability, manufacturability, and competition readiness.
Optimization	Developed a Python script to optimize tail moment, cargo configuration, planform selection, and performance analysis, reducing testing time and improving scoring strategy.
Hands-on Execution	Performed aircraft assembly, structural repairs, and cross-system troubleshooting during build and competition preparation to improve flight stability and operational reliability.
Competition Impact	Contributed to a team outcome of 3rd Overall Worldwide in both Micro and Regular classes at SAE Aero Design East 2022.

Suggested Resume/Website Bullets

- Led the Structure & Stability team, mentoring six members while driving aircraft structural design, assembly, and troubleshooting to improve stability and overall performance.
- Developed a Python optimization script for tail moment, cargo configuration, planform, and aircraft performance analysis, reducing testing time and supporting scoring decisions.
- Led aircraft structural design and improvement initiatives through hands-on assembly, structural repairs, and cross-system troubleshooting to enhance performance, stability, and reliability.

4. Technical Story for Website



Competition aircraft in flight, showing the final airframe configuration.

Problem Statement

The team had to design aircraft configurations that could satisfy competition requirements while maximizing mission performance, scoring potential, structural reliability, and stability during real flight conditions.

Engineering Approach

- Balanced structural integrity with lightweight aircraft design requirements.
- Used stability and performance calculations to guide tail moment, planform, and payload/cargo configuration decisions.
- Translated analysis into build-ready aircraft structures through hands-on fabrication, assembly, and iterative repair.
- Supported competition readiness by troubleshooting cross-system issues affecting aircraft stability and reliability.

Outcome

The final team performance demonstrated strong design validation and execution: 3rd Overall Worldwide in both Regular and Micro classes, including award-winning design reports and mission performance placements.

5. Website-Ready Copy

Hero Section

DJS Skylark - SAE Aero Design Competition

Led structural design, stability analysis, and aircraft build support for an international RC aircraft competition team that secured 3rd Overall Worldwide in both Micro and Regular classes at SAE Aero Design East 2022.

Short Project Description

As part of DJS Skylark, I led the Structure & Stability team and supported the design, optimization, assembly, repair, and troubleshooting of RC aircraft for SAE Aero Design East 2022 in Fort Worth, Texas. I worked on structural design improvements, aircraft stability, and Python-based optimization for tail moment, cargo configuration, planform, and performance analysis. The team competed against 42 teams worldwide and secured 3rd Overall Worldwide in both Regular Class and Micro Class.

Impact Metrics / Counters

42+ Global Teams	3rd Overall Worldwide - Micro	3rd Overall Worldwide - Regular
1st Micro Design Report	6 Members Mentored	Python Optimization Tool

Suggested Tags

Aerospace Design | Aircraft Structures | Stability Analysis | Python Optimization | RC Aircraft | SAE Aero Design | Team Leadership | Troubleshooting | Mechanical Engineering

6. Image Placement Guidance for Codex

Image File	Recommended Website Use
SkylarkLogoColour (1).png	Use in project header or logo badge.
650653982_...webp	Hero image: team photo with aircraft, awards, and Indian flag.
655385107_...webp	Achievement section: award plaques and proof of rankings.
bb6e593f-...png	Technical section: aircraft prototype on ground; use with structural design/assembly copy.
8999bb57-...png	Outcome section: aircraft in flight; use with mission performance and reliability copy.

Suggested Web Layout

- Hero: logo + large team image + one-sentence impact statement.
- Achievement cards: Micro and Regular class results as compact ranking cards.
- Contribution section: 3 cards for Leadership, Structural Design, and Python Optimization.
- Technical story: aircraft images paired with a concise problem-approach-outcome narrative.
- Gallery: competition aircraft, awards, flight photo, and logo.

Sources used: DJS Skylark Team Report 2021-2022; uploaded competition images; Arya Raut resume/project bullets; DJS Skylark 2023 congratulation PDF for broader team context only.